

Temporal Self-Location: A Thought Can Have a History Without Knowing How Long It Has Been Going

Human-readable companion to Temporal Self-Location in Bounded Cognitive Episodes: Separating Process History from Elapsed-Time Access and Control

Human-readable companion to the completed scientific manuscript. This version preserves the scientific claims and limits of the completed manuscript while presenting the motivating problem and argument in more direct language. It does not constitute a separate research contribution.

Abstract

A thought can change many times while it unfolds. It can accumulate evidence, revisit earlier material, generate alternatives, change confidence, and move toward a decision. None of that guarantees that the process has an accessible answer to a different question: how long has this been going? This paper separates the history of a bounded cognitive episode from information that locates the episode in elapsed time. It then separates having an opportunity to obtain precise temporal information from that information having relevance to the policy that decides whether to continue or stop. Later duration reports are not treated as transparent records of what the process represented online, because duration can be reconstructed from remembered content and event structure. The proposed primary experiment crosses precise temporal access with temporal authority in a self-terminated deliberation task. Evidence would have to converge across time-query behavior, continuation or closure, content history, and an unexpected retrospective estimate. The framework is compatible with established timing, memory, prospective-memory, and decision theories; it does not posit a central clock. Its central empirical proposal remains untested.

1. The puzzle: a thought can be busy without knowing where it is in time

Imagine becoming absorbed in a difficult problem. During the episode, the reasoning is not empty. You compare possibilities, reject some, revisit earlier steps, notice new evidence, change confidence, and perhaps arrive somewhere very different from where you began. Then someone asks, without warning, “How long has that been going?” The richness of the internal history does not guarantee an immediate answer.

The scientific point is narrower than the everyday feeling of losing track of time. Every cognitive process unfolds in physical time, and the nervous system contains many timing mechanisms. The question is whether the history that produced the current state also supplies a usable coordinate that locates the ongoing episode in elapsed time.

RepET (Represented Elapsed Time) calls that usable coordinate temporal self-location. In this paper the phrase does not mean mentally traveling into one’s past or future. It means online information that locates the current bounded episode relative to elapsed time or a containing temporal frame. The first distinction is therefore simple: process history is not the same object as temporal self-location.

process history $\neq$ temporal self-location

The reverse dissociation is equally useful. A person can look at a clock and acquire a precise elapsed-time value while knowing very little about the detailed sequence of mental states that occupied that interval. Clock time can locate the episode without reconstructing the episode’s internal history. The two objects often correlate in ordinary life, but correlation is not identity.

2. Why a later duration answer is not enough

A later answer to “how long did that take?” is tempting evidence that the person knew the duration while the episode was unfolding. Timing research gives strong reasons not to make that inference. Prospective timing, where people know in advance that time matters, differs from retrospective timing, where duration is requested unexpectedly. Retrospective judgments can depend on remembered content, event boundaries, segmentation, and reconstruction (Block & Zakay, 1997; Boltz, 1995; Zakay et al., 1994; Yue et al., 2026).

That means an accurate duration report can be produced by more than one route. The episode might have maintained an online temporal coordinate. Or the participant might construct an estimate afterward from how much happened, how many boundaries are remembered, the apparent density of events, contextual knowledge, or other memory cues. A good answer at the end does not by itself identify which route produced it.

This is why the manuscript keeps several objects separate: physical elapsed duration; the sequence of internal states; remembered event structure; a later duration estimate; access to a precise clock-like channel; actual sampling of that channel; the obtained elapsed-time value; and whether time mattered to continuation or closure. Treating all of these as one thing called “time awareness” would erase the inference the experiment is meant to test.

Question	What it establishes	What it does not establish
How long did the episode physically last?	Observer-measured elapsed duration	What temporal information the episode represented or used
What happened inside the episode?	Process or event history	An online metric coordinate of elapsed time
Can precise elapsed time be queried?	Opportunity for precise temporal access	That the channel was sampled
Was the channel sampled?	Acquisition of a temporal value at that moment	That the value governed continuation or stopping
Can duration be reported afterward?	A later temporal judgment	Continuous online temporal indexing
Did elapsed time alter continue/stop policy?	A control-level relation under the task boundary	A unique internal clock mechanism

3. Sleep is a measurement lesson, not the explanation

The manuscript uses sleep research for a limited reason: sleep makes it unusually obvious that objective state, transition history, subjective experience, report, and later recovery can be measured as different dimensions. Serial-awakening studies can pair physiological state with immediate reports, while sleep microstructure such as cyclic alternating pattern provides an objective axis of transition-rich history (Siclari et al., 2013, 2017, 2018; Terzano et al., 1988, 2002; Parrino et al., 2012).

This does not imply that waking deliberation is sleep-like, that ordinary absorption is a sleep state, or that sleep proves the waking hypothesis. It supplies a measurement precedent: a system can have a rich objectively measurable history while reports about experience, duration, recall reach, or richness remain distinct variables. RepET carries that dimensional discipline back to waking cognition.

4. Access to time is not the same as authority of time

The second major distinction concerns control. Precise temporal information can be available without being relevant to what the person should do. A clock can sit in view during a task even when elapsed time has no bearing on whether the best policy is to continue thinking or to stop. Conversely, a task can make time costly, valuable, or deadline-relevant, giving elapsed time a legitimate role in continuation or closure.

RepET calls the first manipulation temporal Access and the second temporal Authority. Access is an opportunity condition, not proof that anyone actually looked. Authority is task-level policy relevance, not a hidden internal module or a generic control weight. The experimental chain is deliberately more careful:

precise channel available ($X=1$) $\rightarrow$ sampled ($Q=1$) $\rightarrow$ elapsed value acquired (T_q) $\rightarrow$ possible policy use

Each arrow needs evidence. $X=1$ means the participant could obtain a precise elapsed-time value. A query shows acquisition at that moment. Acquisition still does not prove that the returned value controlled the next decision. And even evidence of policy use would not identify one unique biological timing mechanism.

The distinction has close precedent. Time-based prospective-memory research already shows strategic clock monitoring, sensitivity to clock availability, and changes in checking behavior as temporal targets and consequences change (Oksanen et al., 2014; Mioni & Stablum, 2014; Mioni et al., 2020; Munaretto et al., 2022; Waldum & Sahakyan, 2013; Huang et al., 2014). RepET does not claim discovery of strategic time monitoring. Its narrower

proposal is to independently cross access to a precise temporal channel with whether elapsed time has genuine relevance to continuation or closure inside a self-terminated deliberative episode.

5. The primary proposed test: cross Access with Authority

The proposed waking experiment is a 2×2 design. Participants deliberate over a task whose episode ends when they choose to stop. In one factor, a precise elapsed-time channel is either queryable or not. In the other, elapsed time either matters to the optimal continuation policy or does not. Because the factors are crossed, the experiment can ask not merely whether a clock changes behavior, but whether temporal information is selectively recruited when it should matter.

Condition	Precise temporal channel	Time relevant to continue/stop policy	Critical interpretation
Access – / Authority –	No	No	Baseline: closure should be explainable without precise external time
Access + / Authority –	Yes	No	Tests whether mere clock/query availability perturbs closure or only supports optional reporting
Access – / Authority +	No	Yes	Tests how behavior adapts when time matters but precise external access is unavailable
Access + / Authority +	Yes	Yes	Critical condition for authority-dependent temporal sampling and time-sensitive closure

The strongest result would not be one number. It would be a convergent pattern: when time has authority and a precise channel is available, participants should alter whether or when they query it, acquire values that are relevant to the continuation problem, and show the corresponding change in closure behavior. Content and event history must be measured because they provide a competing route to duration knowledge. An unexpected retrospective estimate is secondary evidence, not a privileged window into what the online process represented.

The design also separates the causal effect of the query act from the informational value it returns only imperfectly. A self-initiated query can itself become a metacognitive checkpoint, attentional reset, or strategy reconsideration. The current study therefore cannot treat a local query-to-stop correlation as proof that the displayed time caused stopping. Future matched nontemporal queries or perturbations of the returned time value could address that ambiguity.

6. What would count as evidence—and what would not

RepET is an identification proposal, so the manuscript is explicit about weak evidence. Accurate retrospective duration alone is insufficient because reconstruction can be accurate. Earlier stopping under time pressure alone is insufficient because urgency, reward pressure, generic motor preparation, or evidence dynamics can produce the same endpoint. Looking at a clock more often establishes altered monitoring, not necessarily control by the revealed value.

The stronger inference needs the Access $\times$ Authority crossing plus relation-specific measures. When elapsed time has policy relevance, temporal sampling should change selectively, and continuation or closure should change in a way consistent with the task's externally specified payoff structure. Content-history alternatives and generic-pressure accounts must remain competitors. If they explain the pattern equally well, temporal self-location has not been identified as the operative relation.

The experiment also does not establish continuous temporal maintenance between queries. A participant might construct temporal self-location just in time when a query is made. That remains a live interpretation rather than something the design is allowed to erase.

7. Report, Recommend, and Control are different functional roles

A downstream track in the paper asks a separate question: what role can temporal information play once it is available? The framework distinguishes three functions without claiming that they form a developmental or mandatory serial sequence.

- **REPORT:** the person can state or estimate temporal information.
- **RECOMMEND:** temporal information can guide an explicit judgment about what should happen next, such as advising another episode to continue or stop.
- **CONTROL:** temporal information actually changes the person's own continuation or closure behavior.

The point is functional decomposition. Information can be reportable without changing action, can support advice without governing one's current behavior, or can alter control without being accompanied by a verbal thought such as "seven seconds have passed." RepET therefore does not equate conscious time talk with temporal authority.

8. Six ways the stronger proposal can fail

The manuscript treats failure conditions as part of the result. The framework should weaken rather than invent hidden timing machinery when its discriminations do not survive.

Failure condition	What it means
Access changes closure when time has no authority	The clock/query manipulation is not neutral; cueing or interface effects may be sufficient.
Authority fails to change policy	The manipulation did not establish that elapsed time was behaviorally relevant, so null monitoring effects are uninterpretable.
No authority-related change in temporal sampling	A closure effect could be generic urgency or another mechanism rather than active recruitment of precise temporal information.
Content/event history fully explains timing effects	Event-history reconstruction wins for that dataset.
Clock looking improves estimates but not closure selectively	The result supports ordinary external-cue use, not temporal authority over control.
Access and Authority collapse into one salience/pressure factor	The central operational separation fails and must not be rescued with unmeasured hidden variables.

9. What the paper establishes—and what remains open

The conceptual result is that objective duration, process history, remembered event structure, online temporal representation, later temporal report, and time-sensitive control are different measurement objects. A bounded episode can be embedded in a world where an experimenter can measure exact elapsed time without the local cognitive process maintaining or using that coordinate.

The paper does not claim a central internal clock, does not deny endogenous timing, does not identify what temporal information feels like, and does not specify one neural implementation of authority. The Access manipulation removes or supplies a precise external query opportunity; it does not erase internal timing signals. The Authority manipulation makes time normatively relevant to the task; it does not prove a latent authority module.

The primary empirical claim remains proposed and undemonstrated. A positive result would show conditional recruitment of temporal information under an experimentally specified authority relation. It would not show that temporal indexing is absent in ordinary thought, that all cognition follows the same architecture, or that any clinical population has a special timing deficit.

10. Conclusion

A cognitive episode can contain a long chain of internal change without that chain functioning as a metric answer to "where am I in elapsed time?" Later duration reports cannot settle the issue because reconstruction can use event history. Exact clock access cannot settle it because availability is not the same as sampling, and sampling is not the same as control.

RepET therefore turns the puzzle into a testable separation. Give or withhold access to a precise temporal channel. Independently make elapsed time relevant or irrelevant to continuation and closure. Measure whether temporal information is selectively recruited, whether closure changes accordingly, and whether event-history alternatives explain the same result. If those distinctions collapse, the stronger proposal should collapse with them.

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